Soham Karpe, M. Eng.

Machine Vision Engineer / AI Systems Engineer

+49 15510686539 karpesoham@gmail.com

Steinweg 17, 93413 Cham, Germany | Willing to relocate

Date of Birth: 03.02.1998, India Driving License: Class B Portfolio

PROFILE

AI Systems Engineer specializing in **industrial machine vision, real-time image processing, and automated inspection systems**. Experienced in developing **C++ and Python-based image processing pipelines**, integrating **industrial cameras, lighting systems, and embedded hardware** for high-performance production environments. Strong background in **OpenCV, deep learning, and industrial automation**, with hands-on experience building **robust visual inspection solutions from hardware integration to AI inference and HMI control**.

PROFESSIONAL EXPERIENCE

Machine Vision Engineer / AI Systems Engineer (Intern) Jul 2024 – Present | Roding, Germany

• Developed industrial machine vision pipelines using C++ and Python with OpenCV for automated defect detection and visual inspection.
• Integrated industrial camera systems, lighting setup, and image acquisition pipelines for real-time inspection and analysis.
• Optimized image processing pipelines for real-time performance, reliability, and robustness in industrial environments.
• Implemented communication between AI inspection systems and external devices using automation protocols and structured data interfaces.
• Collaborated with software, hardware, and automation teams to integrate machine vision systems into production environments.
• Developed and tested AI-based visual inspection systems combining classical image processing and deep learning models.
• Supported testing, debugging, and validation of machine vision components during system integration and deployment.

Junior Software Developer (Werkstudent) Mar 2024 – Jun 2024 | Roding, Germany
mech-tron GmbH & Co. KG

• Developed anomaly detection pipelines using PyTorch and TensorFlow.
• Built a facial recognition system integrating computer vision models with Python backend services and C++.
• Improved ML workflow reliability through data validation, automation, and edge optimization.

Software Engineer Jul 2020 – Jul 2022 | Pune, India
Evoka Technologies Pvt. Ltd.

• Contributed to software development and early product prototyping during the company's pre-incorporation phase.
• Worked as a Software Engineer post-incorporation on internal projects, contributing to development, testing, and module improvements.
• Gained hands-on experience in a startup environment, building product thinking skills and working with agile practices.

Internship & Bachelor Thesis Sep 2019 – Jun 2020 | Pune, India
Profiroll Technologies India Pvt. Ltd.

• Designed and optimized a compact hydraulic straightening machine to improve mechanical alignment and operational efficiency.
• Performed system modeling and engineering analysis to evaluate mechanical performance and structural stability.

EDUCATION

M.Eng. Artificial Intelligence for Smart Sensors and Actuators Oct 2022 – Mar 2026 | Cham, Germany
Technische Hochschule Deggendorf (Deggendorf Institute of Technology- DIT)

Artificial Intelligence, Model Based Function Engineering, Machine Learning and Deep Learning, Robotics and Autonomous Systems, Big Data, Computer Vision, System Design and Intercommunication, Embedded Systems.

Master Thesis: mech-tron GmbH & Co. KG, Roding
Design and Development of a Real-Time AI-Based Visual Inspection System (Best Thesis Nominated)

AI Visual Inspection • Computer Vision Pipelines • Camera Hardware Integration • Qt-based HMI Development • Industrial Automation • Image Processing with deep learning • Defect Detection • Camera Calibration • Model Training & HMI Configuration • Real-Time Industrial Vision System

Bachelor of Engineering Jul 2016 – Jun 2020 | Pune, India
Pune University, India
Average Grade: 2.3
Programming Fundamentals, Engineering Mathematics, Data Structures (Fundamentals), Control Systems, Mechatronics Systems, FEA, Applied Mechanics, Strength of Materials, Dynamics of Machinery

PROJECTS

Liveness Detection using Computer Vision (Image Processing) Apr 2024 – Jul 2024 | DIT, HiWi
- Developed and evaluated a YOLOv8-based computer vision model for human liveness detection, designing the custom dataset, training pipeline, and experimental setup. Achieved 91% accuracy through systematic model tuning, data augmentation, and performance benchmarking across varied real-world conditions.

Building Height Detection using Advanced Intelligence System Mar 2023 – Jul 2023 | DIT, HiWi
- Developed MATLAB and Python machine learning models for building height estimation, designing feature extraction, visualization, and real-time processing workflows to enable accurate analysis for architectural and urban scenarios.

Pick-and-Place Scara Robot using Gazebo Simulation Mar 2024 – Jul 2024 | DIT, Case Study
- Designed and simulated a pick-and-place SCARA robot in Gazebo using ROS, implementing kinematics, motion planning, and control algorithms to enable precise object manipulation for industrial automation scenarios.

KNOWLEDGE AND TECHNICAL SKILLS

Programming Languages
C++, C, Python, Java

Embedded & GUI Development
Embedded Programming, Qt Framework

Data Tools
Tools & Platforms: Docker, Git/GitHub, CI/CD, Python Automation Scripts
Cloud: Google Cloud (BigQuery), AWS (Basics)

Languages
German- B1 (actively improving, goal C1)
English - B2
Hindi & Marathi- Native

Prompt Engineering
LLM Integration, Prompt Engineering, AI-assisted Workflows

Embedded / Edge Deployment:
Raspberry Pi, ESP32, GPIO, MQTT, Edge Inference, Arduino Mega, Daheng imaging camera, Basler Camera

Productivity Tools & Visualization
Visual Studio Code, Git/GitHub, Linux, PowerShell, QEMU, Matplotlib, Seaborn

Machine Vision & AI
OpenCV, Image Processing, Industrial Cameras, Image Acquisition, Lighting & Exposure Optimization, Defect Detection, Real-Time Vision Systems, Computer Vision, YOLOv8, PyTorch, TensorFlow, scikit-learn, Dataset Preparation, Model Optimization, Real-Time Inference, Hyperparameter Tuning, Industrial Vision Systems.

Soft Skills
Analytical Thinking, Problem Solving, Communication, Team Collaboration, Initiative

CERTIFICATION

Machine Learning with Python • **Introduction to Deep Learning & Neural Networks with Keras** • **AWS Certified Machine Learning - Specialty** • **AWS AI Practitioner Certification Prep** • **Goethe-Institut A2 and B1 (German Super Intensive Course)** • **: Pursuing -** | **Machine Learning Specialization (Andrew Ng)** | **Computer Vision with TensorFlow**

INTERESTS

Strong interest in AI-driven computer vision, industrial inspection systems, and machine learning for automation. Passionate about building real-time visual inspection pipelines and scalable AI solutions for intelligent production systems.

Cham, 12.03.2026